Tutorial Series No. 01

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Exercise 1. The Need for Parallelism (Conceptual)

1. Define: Transistor, Integrated Circuit (IC), Microcontroller, Microprocessor.
2. What is **lithography**, and how does it relate to Moore's Law and Dennard Scaling?
3. What do you know about **Moore's Law** and **Dennard Scaling**? How do they relate to the evolution of computing hardware, and what happened to each one around 2005?
4. What is the relationship between **CPU clock speed**, heat dissipation, and noise? How does this relate to the **power wall**?
5. What is the relationship between **CPU clock speed**, **Memory Frequency** and the **Front-Side Bus (FSB)**? how do they relate to the **memory wall**?
6. What do you know about **Front-Side Bus (FSB)**, and **QuickPath Interconnect (QPI)**? How does QPI relate to the **memory wall**?
7. The **8086** CPU had a single execution unit (Arithmetic Logic Unit, **ALU**) and could execute only one instruction at a time.
   - Is this still true for modern single-core CPUs? Explain your answer with examples of modern single-core CPUs.
8. What is **ILP: Instruction Level Parallelism**, and what is the ILP wall? How does it relate to the **memory wall**?
9. Around 2005, three "walls" stopped single-core scaling. Name them and explain each in one or two sentences. Which one is considered the most important, and why?

Exercise 2. Basic Performance Metrics

1. Distinguish between **wall-clock time**, **latency**, and **throughput**.
2. A sequential program takes $T_1 = 120$ s. Parallelized and run on $T = 8$ workers, it takes $T_T = 30$ s.
   a. Calculate the Speedup, $S(T) = \frac{T_1}{T_T}$.
   b. Calculate the Efficiency, $E(T) = \frac{S(T)}{T}$, and interpret it.
3. What unit is used to express the peak performance of a system in terms of floating-point operations? What do the prefixes Giga, Tera, and Peta stand for?
4. "With 8 cores, will my program run 8 times faster?" Answer in two sentences.

Exercise 3. Applying Amdahl's Law (Fixed Problem Size)

1. Which of these Amdahl's Law formulas is correct? Explain your answer:

$$S(N) = \frac{1}{S+\frac{P}{N}} \qquad \text{or} \qquad S(N) = \frac{1}{(1-P)+\frac{P}{N}}$$

2. Consider a program with $S = 0.15$ (15%). The remaining part is perfectly parallelizable.
   1. Calculate the theoretical speedup for $N = 4$.
   2. Calculate the theoretical speedup for $N = 32$.
   3. Determine the maximum theoretical speedup ($N \to \infty$).
   4. Compute the efficiency for $N = 4$ and $N = 32$. What do you observe?
   5. How many processors are needed to reach a speedup of 5?
   6. What can you conclude about the limitations imposed by Amdahl's Law?
   7. Why is Amdahl's Law often considered "pessimistic"?

Exercise 4. Gustafson's Law (Scaled Problem Size)

1. Which of these Gustafson's Law formulas is correct? Explain your answer:

$$S(N) = S + P \times N \quad \text{or} \quad \mathrm{S}(N) = N - S \times (N - 1)$$

2. Consider $S = 0.1$ and $P = 0.9$.
    1. Show that the two forms of the formula above are equivalent.
    2. Calculate the speedup for $N = 16$ and $N = 64$.
    3. Compute the Amdahl speedup for the same S and N, and compare the results in a table.
    4. Discuss the implications of each law for parallel computing performance.

Exercise 5. Amdahl vs Gustafson on the Same Program

A program run on $N = 16$ processors spends 15% of its total parallel execution time on sequential tasks ($S = 0.15$).

1. Use Gustafson's Law to calculate the scaled speedup.
2. Apply Amdahl's Law with the same S and N and compare.
3. Explain the key difference in assumption between the two laws regarding problem size.
4. The two laws use the same symbol S. Is it measured the same way? Explain.

Exercise 6. Flynn's Taxonomy

1. In the lecture slides, we saw a diagram of Flynn's taxonomy. Four categories were defined:
    - **SISD**: Single Instruction, Single Data
    - **SIMD**: Single Instruction, Multiple Data
    - **MISD**: Multiple Instruction, Single Data
    - **MIMD**: Multiple Instruction, Multiple Data
    1. What do we mean by "Instruction Stream" and "Data Stream" in this context?
    2. The two least used categories share a common characteristic. What is it? why it is either obsolete or rarely used in practice?
    3. How do shared and distributed memory systems fit into Flynn's taxonomy?
    4. How do Data Parallelism and Task Parallelism fit into Flynn's taxonomy?
2. In 1972, Flynn proposed three types of parallel architectures for the SIMD: **array processors**, **pipelined processors**, and **associative processors**. Briefly describe each one with an example of a modern system that uses it.
3. What is the difference between **SIMD** and **SIMT**? Give an example of a modern system that uses SIMT.
4. Classify the following systems into one of Flynn's categories:
    - A single-core CPU with a single ALU and a single FPU.
    - A GPU with thousands of cores executing the same instruction on different data.
    - A cluster of multi-core CPUs, each running different tasks on different data.
    - A vector processor that applies the same operation to multiple data elements in a pipeline.
    - A multi-core CPU with multiple threads executing different instructions on different data.

Exercise 7. Categories of HPC Systems

1. Briefly differentiate **shared-memory**, **distributed-memory**, and **accelerated** systems.
2. Within shared memory, what is the difference between **SMP** and **NUMA**? Why does data placement matter in NUMA?
3. Classify each system into one category: (a) a laptop with an 8-core CPU; (b) three servers linked by a network; (c) a workstation with a CPU and a GPU connected by PCIe.
4. A GPU-accelerated cluster node: classify it with Flynn's taxonomy (a) at the node level, and (b) inside the GPU cores.

Exercise 8. Programming Models and Architectures

Match each programming model with its system category and its way of exchanging data.

Programming Model	System Category	Data exchange
OpenMP	Shared-Memory Systems	.....
MPI	.....	.....
CUDA / OpenCL	.....	.....
Pthreads	.....	.....

Exercise 9. Synthesis, Anatomy of a Supercomputer

A supercomputer has 1000 nodes. Each node has 2 CPU sockets (multi-core) and 4 GPUs.

1. Describe the system as layers (cluster, node, core, accelerator) with the memory model of each.
2. Which programming model would you use at each layer?
3. Why can this machine be described as “a MIMD machine with SIMD inside”?